

SAVINDI WIJENAYAKA, Ph.D.

Senior Machine Learning Engineer & Researcher

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Summary

Machine Learning Engineer with a PhD and 6+ years of applied ML experience across industry and research, developing and deploying production-grade deep learning systems in cloud-native environments. Brings technical depth across the entire journey from problem framing and dataset construction through model architecture, training, and system design, to production deployment, owning each phase with a focus on business impact, engineering rigour, and data governance. Driven by the challenge of turning rigorous research into systems that create tangible, real-world impact across diverse domains and data types.

Technical Skills

- **ML & Research:** Computer Vision, NLP, Signal Processing, Biological modelling and computational analysis
- **Languages & Frameworks:** Python, PyTorch, Keras, Flask, Django, Java, Spring Boot
- **Data types:** Unstructured: Text, Image, Video, Audio, 3D, Point Cloud, Mesh; Structured: Time Series, Tabular
- **Data & Cloud Infrastructure:** Azure, AWS, Docker, Kubernetes, Data warehouse (Snowflake), Relational (MSSQL), Time series (Azure Data Explorer), Stream data (Azure Stream Analytics), Transact-SQL
- **Engineering:** Git, GitHub Actions, Azure DevOps Pipelines, REST, gRPC, Agile, Claude, Cursor

Experience

Intermediate Machine Learning Engineer

Auror · Full-time

Sept 2025 - Present

Auckland, New Zealand

Auror is a retail crime intelligence platform that provides AI-enhanced capabilities to prevent crime.

- Spearheaded scoping, planning, and architecture of an internal RAG-based compliance questionnaire system, driving cross-functional coordination across teams and directing another engineer through full-stack implementation.
- Designed a multi-gate evaluation framework for Auror's flagship entity resolution product, resolving class imbalance and circular bias; shipped a new model to production via a region-separated CI/CD pipeline.
- Architected a modular ML evaluation and fairness reporting framework for prospect trials, reducing report turnaround by 80% and accelerating the path to technical win.
- Prototyped an LPR camera health monitoring system by mapping vehicle detections to spatial heatmaps and flagging deviations from optimal detection zones, enabling automated identification of camera obstructions.
- Established team-wide ML engineering standards via agentic coding conventions and a data licensing review, forming the foundation for a broader data governance framework.

Machine Learning Engineer

WSO2 · Full-time

Sept 2020 - Nov 2021

Colombo, Sri Lanka

WSO2 is a leading open-source integration vendor; Choreo is its AI-enhanced platform-as-a-service product.

- Researched and delivered a GPT-based model for automated API test case generation, deployed on Azure Kubernetes Service, reducing testing effort for customers and enabling an AI-assisted developer workflow.
- Co-architected a time series-based AI Anomaly Detection system using an event-driven microservices architecture across Azure Stream Analytics, Event Hubs, and Function Apps, enabling customers to detect abnormal application behaviours in real time.
- Implemented a regression testing pipeline for the Ballerina Language Server, systematically surfacing performance degradations and resource utilisation issues to ensure production reliability for users.
- Diagnosed and resolved a critical memory leak in the Ballerina Language Server, applying systematic profiling with JMeter and Eclipse Memory Analyser, to eliminate server crashes and restore resource efficiency.

Software Research Engineer

Pearson · Internship

📅 Sept 2018 - Sept 2019

📍 Colombo, Sri Lanka

Pearson is a leading global education provider offering curriculum materials, multimedia learning tools, and testing programs.

- Researched and implemented deep learning modules for emotion detection and speech analysis, integrating models into scalable RESTful microservices for a real-time AI Public Speaking Evaluator.
- Researched and built a Q&A chatbot for Pearson's educational content, iterating from BiDAF to fine-tuned BERT and GPT-2 models to improve answer quality and enable conversational querying of learning material.
- Investigated and developed a flashcard classification service using ULMFiT, LSTM, and GRU to automatically categorise flashcards, reducing manual tagging effort and improving content discoverability for students.

Education

Ph.D. in Bioengineering

University of Auckland

📅 Dec 2021 - May 2025

📍 Auckland, New Zealand

Spearheaded an end-to-end deep learning system for automated 3D biological structure analysis, owning every stage from dataset construction and model architecture through ablation studies and the final computational pipeline, resulting in a reproducible workflow applied across 20+ distinct biological samples.

- Engineered a semantic segmentation model with multiscale attention mechanisms, conducting extensive ablation studies to optimise architecture and delivering an automated analysis pipeline that reduced manual processing time by 40+ hours per dataset.
- Developed a Python-based 3D quantification framework grounded in advanced mathematical methods, transforming raw micro-CT data into structured geometric benchmarks and standardising the full workflow from ground-truth labelling to downstream modelling and simulation.

B.Sc. (Hons.) in Software Engineering

University of Kelaniya

📅 Feb 2016 - Mar 2020

📍 Kelaniya, Sri Lanka

- Specialised in Data Science and Net-centric application development.
- Attained a GPA of 3.96 out of 4.00, obtaining a First Class.
- Dean's List Honouree, recognised in all four academic years of the B.Sc. programme.

Knowledge Sharing & Technical Outreach

- Member of the teaching team for Code In Place 2021, an online Python course offered by Stanford University, contributing to global tech education initiatives.
- Served as a guest speaker for multiple technical webinars (organised by IEEE and DeepLearning.AI), effectively communicating complex deep learning topics to broader audiences.
- Authored technical articles on Medium covering conceptual topics (CNNs, Kubernetes internals, JVM), practical applications of cloud-native microservices (Kubernetes, Docker, Azure), and automated CI/CD pipelines (GitHub Actions, Azure ARM templates).

Achievements

- **1st Place (2022) and 2nd Place (2024)** in the international SPARC FAIR Codeathon, representing the University of Auckland, organised by the SPARC Data and Resource Centre and the NIH.
- **4th Place** in DataStorm 2020 Datathon, organised by Octave (JKH) and University of Moratuwa.
- **1st Runner-Up** in National Youth Software Competition 2017, organised by UNDP Sri Lanka.
- **Vice President** of Marketing & Communications at AIESEC in the University of Kelaniya in 2018, contributing to local chapter growth.